Since, 3 wholes and 2 tenth is multiplied with 2.1, we split the same area into 2 wholes and 1 tenth along its breadth (*Fig. 1.11*).

Here, each row contains 3 wholes and 2 tenth. Each column contains 2 wholes and 1 tenth. The entire area model represents 6 wholes, 7 tenth and 2 hundredth.

Therfore, $3.2 \times 2.1 = 6.72$

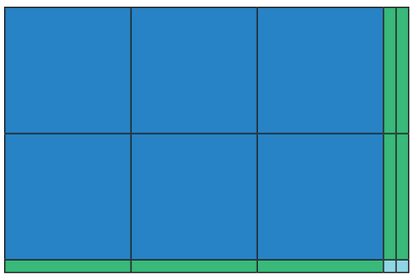


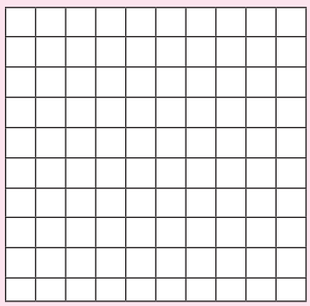
Fig. 1.11

Think

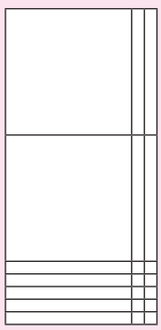
How are the products 2.1×3.2 and 21×32 alike? How are they different.

Try these

(1) Shade the grid to multiply 0.3×0.6.



(2) Use the area model to multiply 1.2×2.5



Example 1.18 Multiply the following

(i) 2.3×1.4 (ii) 5.6×3.2

Solution

(i) 2.3×1.4

First, let us multiply 23×14

$23 \times 14 = 322$

Now $2.3 \times 1.4 = 3.22$

2.3	← 1 decimal place
× 1.4	← 1 decimal place
92	
23	
3.22	← 2 decimal place(s)

(ii) 5.6×3.2

First, let us multiply 56×32

$56 \times 32 = 1792$

Now $5.6 \times 3.2 = 17.92$

5.6	← 1 decimal place
× 3.2	← 1 decimal place
112	
168	
17.92	← 2 decimal place(s)

Example 1.19 Latha purchased a churidhar material of 3.75 *m* at the rate of ₹62.50 per metre. Find the amount to be paid.

Solution

Cost of churidhar material = ₹62.50 per metre

Length of churidhar material = 3.75 *m*

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